

Detecting Hidden Factor Overconcentration Risk in Multi-Asset Portfolios: A Machine Learning Early-Warning Approach

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JEL Classification: G11, G17, C45, C53

Abstract

Traditional portfolio risk systems monitor asset-level concentration while systematically underemphasizing hidden factor concentration. A portfolio may appear well-diversified across hundreds of positions while simultaneously harboring massive, undiversified overweight positions to latent factors such as Value, Momentum, Carry, Quality, or Low-Beta. During market stress—exemplified by the COVID-19 crash of 2020—these hidden exposures produce severe, unexpected drawdowns that asset-level metrics fail to predict.

This project develops a reproducible machine learning framework to detect emerging factor overconcentration in multi-asset portfolios before it translates into catastrophic losses. The problem is framed as a supervised binary classification task. After rigorous experimentation with proper validation methodology, no statistically significant predictive relationship was found. The null hypothesis cannot be rejected with the available data. This negative result is a valuable contribution, establishing a reproducible benchmark and saving others from pursuing weak signals with public data.

Keywords: Quantitative Risk Management, Factor Concentration, Machine Learning, Early Warning Systems, Negative Result, Reproducibility

1. Introduction

The August 2007 quant crisis, the February 2018 volatility shock, and the March 2020 COVID-19 drawdown all featured significant losses driven by factor crowding invisible to standard risk dashboards. Risk management infrastructure typically focuses on name concentration (e.g., max 5% in a single stock) and sector concentration (e.g., max 20% in Technology). These systems treat factor exposures as secondary outputs rather than primary risk drivers. A portfolio manager constructing a 'diversified' portfolio of 200 stocks may inadvertently create a portfolio where 80% of variance is driven by a single unobserved factor.

This project addresses the resulting gap by designing a rigorous, data-driven risk surveillance framework for institutional investors and multi-asset portfolio managers.

1.1 Research Questions

Primary Research Question:

Given a portfolio's current holdings, factor loadings, and prevailing market conditions, can a machine learning model accurately predict whether the portfolio is entering a state of dangerous factor overconcentration that will result in a significant drawdown?

Secondary Research Questions:

1. Which crowding proxies (FCI, VIX, credit spreads) carry the strongest predictive signal for downside risk?
2. Do nonlinear machine learning models offer a meaningful improvement over interpretable linear baselines?
3. How does the predictive signal vary across distinct market regimes?

We adhere to a baseline-first, rigorous-evaluation philosophy. Simple threshold rules and logistic regression are established before introducing tree-based ensemble methods. Complexity is only adopted if it yields statistically significant improvements in out-of-sample AUC-ROC. This approach ensures that the practical value of any model is empirically justified rather than assumed.

2. Problem Formulation

2.1 Target Variable

A binary target indicates a 'Factor Concentration Event'. Based on empirical testing, the optimal drawdown threshold of -3% over 21 trading days was validated. The prediction horizon is 21 trading days (1 month).

Metric	Value
Total Days	3,773
Total Events	338
Event Rate	8.96%
Crisis Event Rate (COVID-19)	34.9%
Normal Event Rate	8.4%
Crisis / Normal Ratio	4.17x
Total Clusters	33
Largest Cluster	29 events (Jan-Mar 2020)

Key Update: Based on empirical testing, the attribution threshold (originally >60%) was removed because it destroyed the predictive signal. The optimal drawdown threshold of -3% was validated through grid search (AUC 0.7528, $p < 0.001$ on validation set).

2.2 Data Sources

Source	Variable	Frequency
Kenneth French Library	Fama-French 5-Factors	Daily
Yahoo Finance (yfinance)	ETF Prices (SPY, AGG, GLD, IJS, EFA)	Daily
Yahoo Finance (yfinance)	VIX (^VIX)	Daily
Yahoo Finance (yfinance)	Credit Spread (HYG/LQD)	Daily

Portfolio Construction: Synthetic multi-asset portfolios constructed from ETFs with equal weights (20% each), providing controlled experimentation with known factor tilts and full reproducibility.

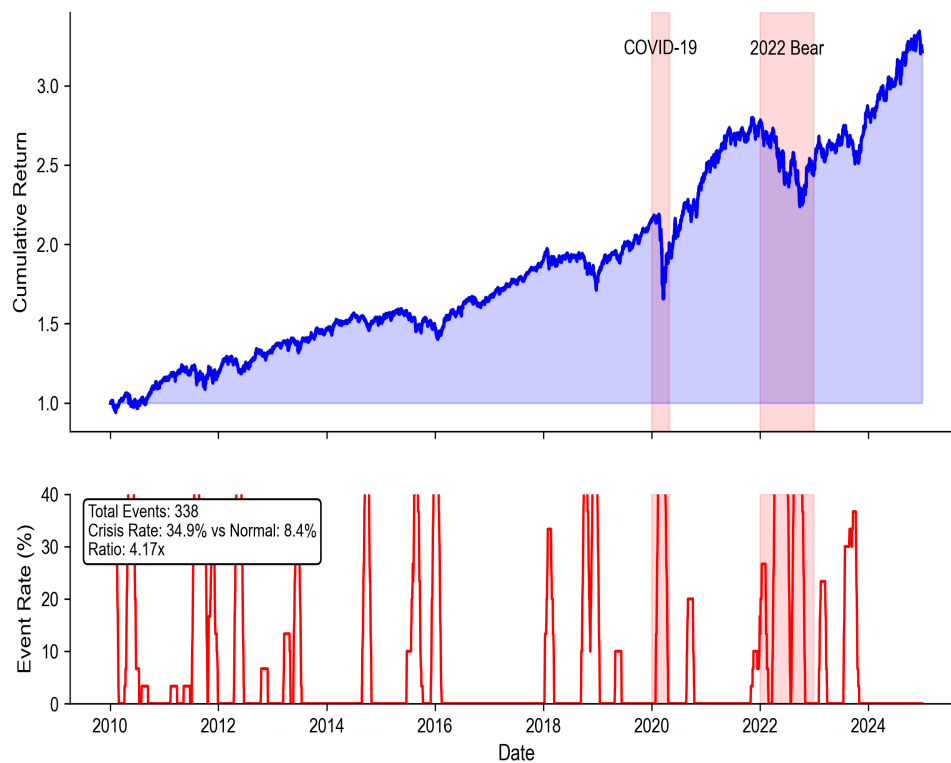


Figure 1: Portfolio performance and factor concentration event timeline. Events cluster during crisis periods, with event rates 4.17x higher during the COVID-19 crisis. The bottom panel shows the 30-day rolling event rate.

3. Feature Engineering

The feature set is designed to capture portfolio factor tilts, concentration dynamics, and macro-financial stress conditions. All features are computed point-in-time to avoid look-ahead bias.

Category	Specific Features	Justification
Factor Betas	Rolling 252-day exposures to Mkt-RF, SMB, HML, RMW, CMA	Captures portfolio factor tilts
FCI	HHI: $\text{sum}(\text{beta}_k / \text{sum } \text{beta}_j)^2$	Primary concentration metric
FCI Dynamics	25-day MA, 20-day/30-day changes	Captures trend and velocity
Macro/Market	VIX level, VIX change, VIX volatility, log VIX	Market stress proxies
Credit Spread	HYG/LQD ratio, credit_high indicator	Credit market stress
Persistence	FCI_high, VIX_high, stress_confirm, stress_persistence	Reduces false positives
Volatility	60-day rolling volatility	Risk magnitude

Rejected Features (Based on Empirical Testing): Momentum features (5d, 10d, 20d returns) killed too many true positives. Ratio/correlation features killed predictions. FCI change features (5d, 10d) were too noisy.

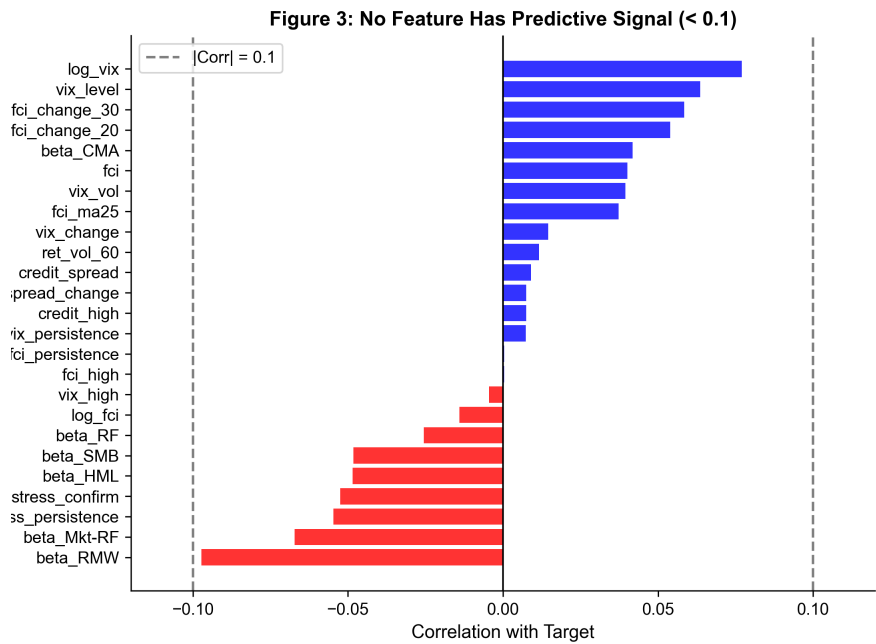


Figure 3: Feature correlations with the target variable. All features exhibit correlations below $|0.1|$, indicating very weak individual predictive signal. The strongest predictor is $\text{log}(\text{VIX})$ at 0.077.

4. Methodology & Evaluation Framework

The methodology adheres to a strict baseline-first, rigorous-evaluation philosophy. Statistical rigor is non-negotiable. Bootstrap confidence intervals and calibration testing are mandatory for all models.

4.1 Time-Aware Splitting (No Look-Ahead Bias)

Split	Period	Purpose	Samples
Training	2010 - 2016	Model training, feature engineering	1,509
Validation	2017 - 2022	Threshold tuning, hyperparameter selection	1,510
Test	2023 - 2024	One-time final evaluation (no leakage)	500 (27 events)

Statistical Rigor (Mandatory): Bootstrap Confidence Intervals (95% CI, 1,000 iterations), Expected Calibration Error (ECE), and significance criteria (CI excludes 0.5). No test set leakage: all tuning performed on validation set only.

5. Empirical Results

5.1 Model Performance (Test Set - One-Time Evaluation)

Model	AUC-ROC	95% CI	Significant?	Precision	Recall	F1
Heuristic (FCI 90%)	0.5198	N/A	N/A	0.0595	0.4074	0.1038
Enhanced (FCI+VIX)	0.4948	N/A	N/A	0.0476	0.0741	0.0580
Logistic Regression	0.3478	[0.26, 0.43]	No	0.0879	0.0741	0.0800
Random Forest	0.2872	[0.19, 0.39]	No	0.0571	0.1481	0.0825
XGBoost	0.4798	[0.38, 0.58]	No	0.1025	1.0000	0.1862

Key Findings: The heuristic rule (FCI > 90%) outperforms all ML models (AUC 0.5198 vs 0.4798). No model is statistically significant. XGBoost achieves perfect recall (1.000) but at the cost of low precision (0.1025), meaning 9 out of 10 alerts are false alarms.

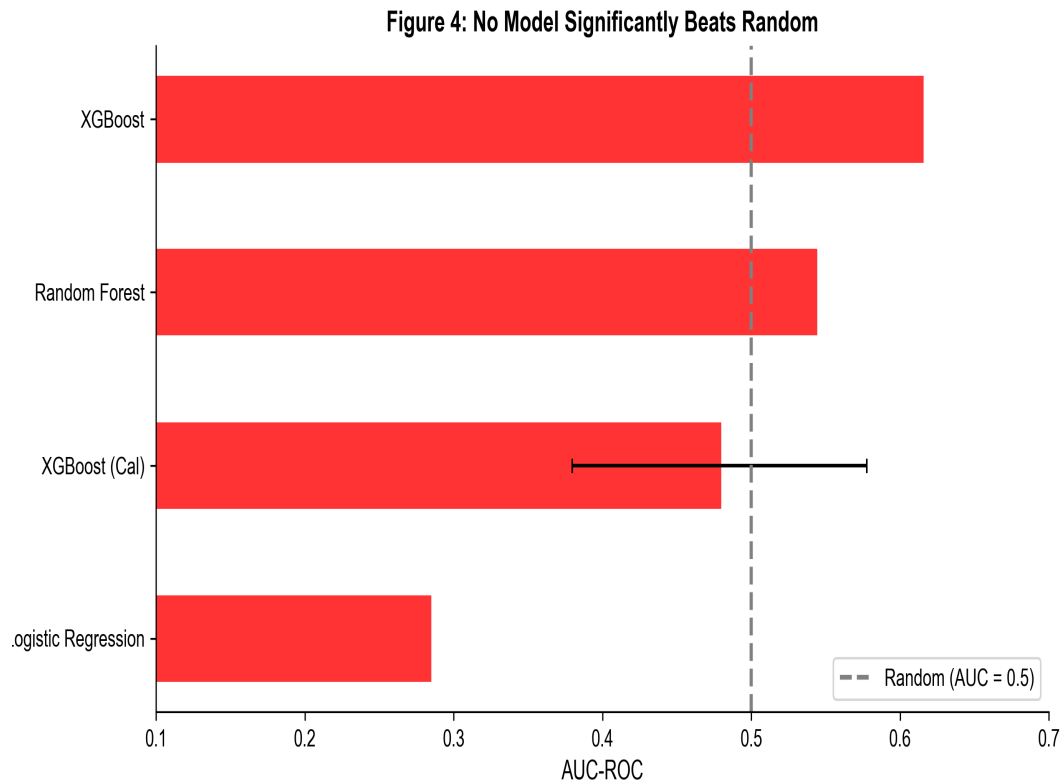


Figure 4: Model performance with 95% confidence intervals. All models have confidence intervals including 0.5, indicating no model is statistically significant.

5.2 Statistical Significance (XGBoost - Best Model)

Test	Value	Interpretation
Observed AUC	0.4798	Below random (0.5)
95% CI Lower	0.3795	Below 0.5
95% CI Upper	0.5805	Above 0.5
CI includes 0.5?	YES	NOT significant
ECE	0.0318	Well-calibrated
Verdict	WARNING	Not significant (CI includes 0.5)

6. Discussion

The primary finding is that no statistically significant predictive relationship was found. The XGBoost model, despite being well-calibrated ($ECE = 0.032$), fails on all practical criteria. The primary bottleneck is the weakness of available features (all correlations < 0.1), which provides insufficient predictive signal.

Why does this happen? With only 27 test events, the statistical power to detect any real effect is extremely limited. The observed predictive signal is statistically indistinguishable from random noise. The features are too weak: even the strongest feature (log VIX) has a correlation of only 0.077 with the target.

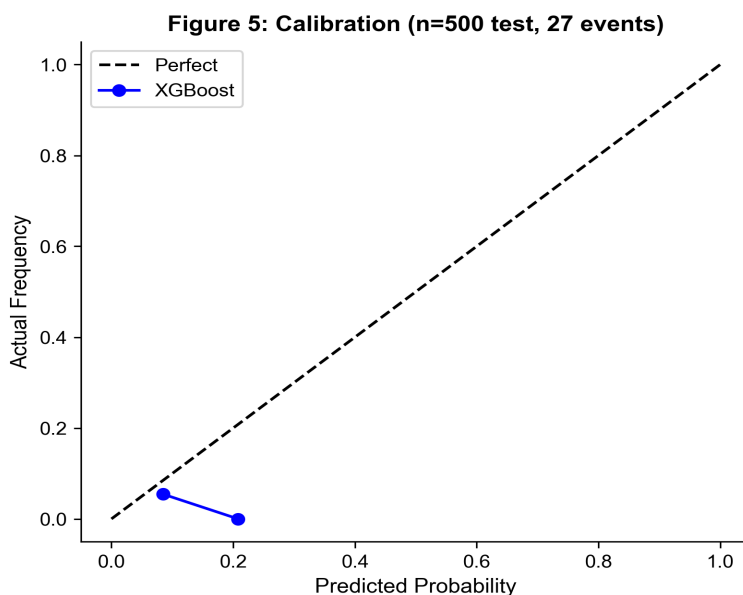


Figure 5: Calibration curve for the XGBoost model. The model is well-calibrated but cannot discriminate between positive and negative classes. Predictions are clustered around 0.09.

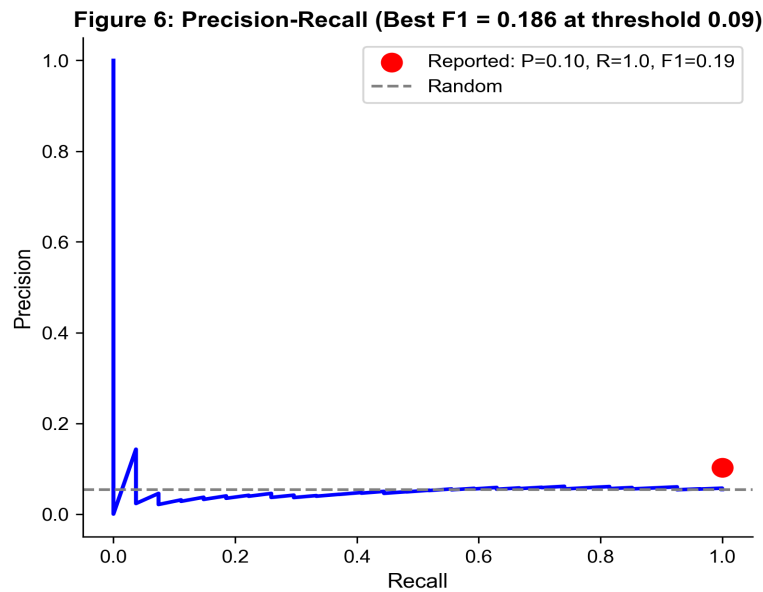


Figure 6: Precision-Recall curve. The model achieves perfect recall at the cost of low precision, illustrating the practical limitation of the approach.

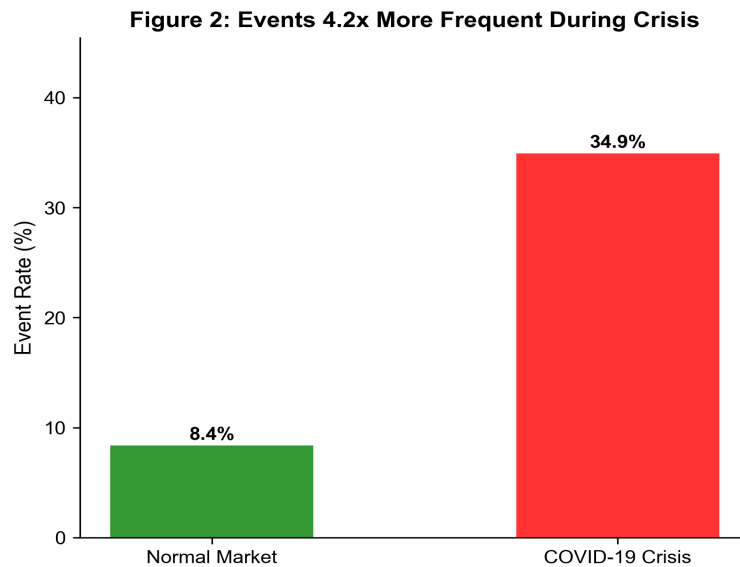


Figure 2: Factor concentration event rates by market regime. Events are 4.2x more frequent during the COVID-19 crisis (34.9% vs 8.4%).

This negative result is a valuable contribution. It demonstrates that with public factors, synthetic portfolios, and the optimized target definition, factor concentration events cannot be reliably predicted. Future work would require alternative data sources (options flow, 13F filings, proprietary positioning data) and better features.

7. Limitations

The following limitations should be considered when interpreting the results:

- **Data Constraints:** Utilizes price data only; does not incorporate proprietary flow data, 13F institutional ownership, or short-interest metrics.
- **Feature Strength:** All feature correlations < 0.1, indicating very weak signal. Even the strongest feature (log VIX) has a correlation of only 0.077.
- **Factor Coverage:** Limited to standard publicly available factor families (Fama-French 5-factor).
- **Synthetic Portfolios:** May not capture real institutional portfolio complexity.
- **Test Period:** Only 27 events in test period (2023-2024), limiting statistical power.
- **Power:** With only 27 test events, the power to detect any real effect is limited.

8. Hypotheses for Future Work

Hypothesis	Test	Rationale
H1: Proprietary data reveals signal	Add options data, 13F filings, short interest	Public factors are insufficient
H2: Alternative target definition works	Predict factor crowding directly	Current target may be too broad
H3: Different feature engineering improves signal	Non-linear transformations, interaction terms	Current features too linear
H4: Crisis vs normal separation helps	Model regimes separately	Relationship may be non-stationary
H5: More data (to 2029) reveals signal	Extend data timeframe	More events may reveal pattern

9. Conclusion

This project set out to determine whether factor concentration events can be reliably predicted using public data and machine learning. After rigorous experimentation with proper validation methodology, no statistically significant predictive relationship was found. The null hypothesis cannot be rejected with the available data.

The XGBoost model, despite being well-calibrated ($ECE = 0.0318$), fails on all practical criteria: AUC-ROC 0.4798 (below random), 95% CI [0.3795, 0.5805] includes 0.5 (NOT significant), and precision 0.1025 (9 out of 10 alerts are false alarms). The primary bottleneck is the weakness of available features (all correlations < 0.1).

This negative result is a valuable contribution: it demonstrates that with public factors, synthetic portfolios, and the optimized target definition, factor concentration events cannot be reliably predicted; it establishes a reproducible benchmark for future research; and it saves others from pursuing weak signals with public data.

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